

Ammar Yasser Mohamed Abd-El Aal

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OBJECTIVE

Aspiring Mechatronics Engineer with experience in robotics, embedded systems, and mechanical design, passionate in R&D and Electric mobility fields.

SUMMARY OF QUALIFICATIONS

- **CAD, CAM, 3D modeling** and stress analysis with 10+ projects that fit the need and catch the eye.
- Great experience with Mechatronic system design stages, system integration, modeling, verification and validation.

EDUCATION

Ain Shams University (ASU)

Cairo, EG |2022 – 2027|

Bachelor of Engineering in Mechatronics department (Undergraduate)

Fields of study: Mechanical design principals – Mechatronic System Design – Control

Masr International Computer Academy (MICA)

Giza, EG |2019 – 2022|

Excelled in computer programming, self-learning and online researching, and score the 4th on the national level with score of 98.2%.

TECHNICAL SKILLS

- **Mechanical design:** SolidWorks, Fusion 360, Inventor, CorelDraw, SketchUp.
- **Robotics:** MATLAB robotics toolbox, SIMULINK, SIMSCAPE multi body.
- **Design for manufacturing concepts (DFM):** Optimized designs for cost, volume, and manufacturability.
- **Micro Controller Programming:** ESP32-S2, AVR Atmega32, Ai Thinker PB-03f.
- **Battery Design:** Simscape, Simulink, Battery Builder.
- **Report writing:** Claude co-work, Latex, Word.

TRAINING & EXPERIENCES

- **Trophy Technologies "Start up" | Tech Team Leader | 2026 June – Present**
Lead the technical team of the R&D department including Embedded System, Software, Simulation Teams. **Battery design**, MATLAB Battery Builder, Esp32-s2 SOC, PCB design, MQTT.
- **KAIRO "Start up" | Mechanical designer | 2026 Jun – 2026 Sep**
Worked on the System Concept design of a portable bicycle-to-electric conversion kit, Analyzed the system on different load studies, power consumption analysis, motor actuation sizing, and power transmission design.
- **ASU Innovation hub Royal Competition | Candidate | 2026 March – 2026 May**
Worked on a detailed initial concept for an auto-packing machine extension (tea packaging) at Royal Herbs Company using SolidWorks, Fusion360 applying DFM principles and optimizing layout for efficiency.
- **Siemens Digital Industry | Aug 2025**
SIMATIC programming 1 in the TIA Portal, Overview and significant performance characteristics of the SIMATIC S7 system family. TIA Portal: STEP7, WinCC, Start Drive.

- **Industry 4.0 Innovation hub, ITIDA | FAB LAB operator | Aug 2025**

Experienced in managing and operating Fab Lab equipment and resources also Standardization documentation Developed and documented operational standards for Fab Lab processes and services.

- **ASU Innovation hub | Trainee | 2025 March**

I have learned all about 3D printing machines, CNC coordination systems, filament types, slicing options and fatal mistakes of 3D printing to avoid.

- **Nawah for Entrepreneurship | Competitor | 2019-2021**

We have won 2nd in 2020 & the 3rd in 2021 on the national level, I learned through my contribution to manage a real-world project and standard workflow approach.

CERTIFICATION

- SolidWorks CAD design Associate | 2025/July | **CSWA, CSWP.**
- Udemy course platform | 2024/Nov. | [Mechanical design via SolidWorks.](#)
- MathWorks platform | 2025/June | **MATLAB, SIMULINK.**